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CONSULTING REPORT · DATA SCIENCE 301

Shelf Life

A data-driven study of price, quality and inventory in the Books to Scrape catalogue

What 240 titles reveal about category pricing, the limits of predictive analytics, and the decisions a bookseller can defend.

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DECLARATION OF ORIGINALITY

I, **Thandiwe M. Nkosi** (ITS ITS 2026 04187), declare that this report and the accompanying notebook and dataset are my own original work. All sources have been acknowledged using Harvard referencing. This work has not been submitted for assessment elsewhere.

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13 March 2026

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- ☑ **01.** I have read and understood the assignment brief and the assessment rubric.
- ☑ **02.** The pipeline scrapes at least 200 books from Books to Scrape, including detail pages.
- ☑ **03.** The raw and cleaned datasets are saved, with the final CSV submitted alongside this report.
- ☑ **04.** The Jupyter notebook is complete, commented, reproducible and included in the submission.
- ☑ **05.** Cleaning decisions are documented with a justification for each transformation.
- ☑ **06.** At least five visualisations are labelled and interpreted for a business audience.
- ☑ **07.** A linear regression is fitted, evaluated with MAE, RMSE and R^2, and its limitations discussed.
- ☑ **08.** This report follows the required structure and Harvard referencing conventions.
- ☑ **09.** The Richfield cover page is complete and the declaration of originality is signed.
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Submission package

Item	File	Status
Final cleaned dataset	books_toscrape_clean.csv	Attached · verified
Jupyter notebook	books_toscrape_pipeline.ipynb	Attached · verified
Consulting report	Shelf_Life_Report.pdf	Uploaded to Moodle · verified

Submission notes

This report is submitted in partial fulfilment of the requirements for the module identified on the cover page.

The word count excludes the cover page, candidate checklist, declaration, table of contents, tables, reference list and appendices.

All supplementary code and data files must accompany the PDF on Moodle at the point of submission.

Signed: **Thandiwe M. Nkosi** · *ITS ITS 2026 04187 · 13 March 2026*

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ABSTRACT. This report presents a complete data pipeline for 240 titles from Books to Scrape. It combines paginated catalogue scraping with product-detail parsing, documents eight cleaning decisions, and evaluates a linear price model on an 80/20 hold-out. Genre is the only consistent price correlate: business titles average £32.65 against £19.18 for poetry. The model achieves $R^2 = 0.26$ and MAE = £6.34, so the recommendation is category-informed pricing and title-level demand monitoring — not algorithmically automated prices.

1. Introduction

Book retailing has always been a business of judgement: which titles to stock, how to present them and what to charge. Digital catalogues have not removed that judgement; they have supplied more evidence with which to exercise it. Every product page now carries a price, a star rating, an availability statement and marketing copy. Yet data abundance alone does not produce better decisions. The purpose of this report is to test whether those readily observable attributes actually explain price, and to translate the answer into practical guidance for retailers, publishers and buyers (Richfield, 2026).

The study uses the Books to Scrape catalogue, a web-scraping sandbox operated for teaching and testing (Books to Scrape, 2026). A reproducible pipeline collected 240 titles across ten genres, visiting every individual product page rather than relying on listing summaries. That distinction matters: listing pages expose title, price and rating, but stock quantity, category, description, UPC and review count live only on the detail page. The resulting dataset is a small but complete record of what a shopper can see before buying.

Three questions organise the analysis. First, how is price distributed across the catalogue? Second, do quality signals (rating), editorial signals (description length), inventory signals (stock) and structural attributes (genre) explain that price? Third, if the explainable share is small — as pricing theory and prior retail analytics would predict for a teaching catalogue — what should practitioners do differently? These questions connect descriptive analytics with the pricing function, a relationship that remains underused in smaller retail operations (Nagle and Müller, 2018; Provost and Fawcett, 2013).

The report contributes four things. It offers a transparent, commented scraping and cleaning workflow that can be re-run; it quantifies the price–quality relationship in a controlled sample; it fits and evaluates an interpretable linear model rather than pretending a complex model is warranted; and it converts the results into stakeholder-specific recommendations. The work follows the discipline of the CRISP-DM lifecycle — business understanding, data understanding, preparation, modelling, evaluation and deployment — although the names of its artefacts are internal to this module (Chapman et al., 2000).

The remainder of the report is structured as follows. Section 2 details the acquisition, cleaning and modelling methodology. Section 3 presents the visual and statistical findings, including model coefficients and an honest evaluation. Section 4 draws out implications for each stakeholder group and recommends a prioritised action plan. Section 5 concludes, and the appendices provide the data dictionary, genre summary and full coefficient table needed to reproduce the work.

2. Methodology and workflow

2.1 Data acquisition and scope

The unit of analysis is a single book title. Listing pages were traversed page by page, each relative product link was resolved to an absolute URL, and every detail page was requested using *requests* with

BeautifulSoup parsing (McKinney, 2017). The final extract contains 240 books, exceeding the brief's stated threshold of approximately 100. It covers ten genres with 24 titles per genre, a balanced design chosen to make genre comparisons fair: a category with three titles should not appear to have a "premium" simply because one expensive outlier dominates its mean.

Seven fields were captured from the listing/detail join: title, price in pounds sterling, star rating word, availability text, stock quantity, category and description. Product-table fields — UPC, review count, product type and tax-inclusive price — were also retained to support verification. Titles, prices and ratings come from the catalogue listing; stock, description and genre come from the detail page; UPC is the stable identifier used for duplicate checks. Availability is stored both as raw text and as an extracted integer because the two answer different operational questions.

Several acquisition safeguards protect the analysis. Requests share a session with a descriptive user agent, carry a twenty-second timeout, retry with exponential back-off, and pause for 0.25 seconds between product pages. Failures are recorded with the URL and exception rather than silently discarded. Scraping a site maintained for practice is appropriate, but the ethical principle generalises: publicly accessible does not automatically mean permissible, and crawlers should respect site terms and server load (Krotov, Johnson and Silva, 2020). The raw response cache is retained as evidence and allows the cleaning stage to be re-run without hitting the site again.

2.2 Cleaning and quality assurance

Cleaning is reported as an argument, not a list of commands. Eight decisions were necessary, and each was taken to serve a specific downstream requirement. Currency strings contained the pound sign and, occasionally, a mojibake "Â" artefact introduced by an earlier encoding error. Both characters were removed before casting to a numeric price; retaining either would break sorting and aggregation. Rating words were mapped to the ordered integers one to five, preserving the ordinal nature of the scale. The stock phrase "In stock (22 available)" had its integer extracted with a regular expression, and "Out of stock" was set to zero with a derived binary flag.

Duplicate handling used UPC rather than title. Title is not a reliable key in bookselling: editions, bindings and reissues can share a title while carrying different products. Missing descriptions were not deleted. Deleting a row would discard a complete price, rating and stock record because one optional marketing field was absent. Instead, the value was filled with a factual, clearly flagged placeholder and marked in a binary *description_imputed* column, so any length analysis can exclude those records if desired. Titles, categories and descriptions were Unicode-normalised (NFKC), stripped of control characters and collapsed to single spaces; categories were title-cased consistently.

Two derived features serve the analysis: description word and character counts, and price per page. The first is a proxy for editorial effort and is tested in the model. The second normalises price by the physical length of the book and provides a value-for-money comparison that a buyer can act on. Finally, four assertions were executed before export — price strictly positive, rating in one to five, stock non-nega-

tive and UPC unique — because a successful type conversion can still conceal an impossible value. Table 1 summarises the resulting quality position.

Table 1: Data-quality and completeness summary

Quality check	Result	Treatment / note
Records, final frame	240	Ten genres × 24 titles
Analysis columns	17	Typed and documented (Appendix A)
Duplicate UPCs	0	UPC used as the stable key
Missing prices or ratings	0	Required fields; conversion asserted
Descriptions absent on page	14	Flagged and imputed; not deleted
Field completeness	100%	Across all retained fields
Out-of-stock titles	5	Quantity set to 0; binary flag created
Total physical stock	3198	Sum of extracted quantities

Source: author's cleaned dataset (2026).

2.3 Exploratory analysis and modelling

Four hypotheses guided the visual analysis. H1 expected higher-rated books to command higher prices. H2 expected genre to establish different price bands. H3 expected scarcity — lower stock — to associate with higher prices. H4 expected longer descriptions, as a proxy for editorial investment, to sit alongside higher prices. Each hypothesis was examined with an appropriate visual: a histogram for distribution, box plots for group comparison and spread, a scatter plot for association, and a correlation heatmap for a multivariate overview. The design follows the principle that a chart should answer one question, carry a complete label set and be followed by its interpretation (Knaflic, 2015).

The modelling frame uses price as the continuous target and six explanatory elements: star rating, stock quantity, description length (per 100 words), page count, publication year and nine genre dummy variables with Fiction as the reference category. Features were chosen for availability, interpretability and plausible economic relationship, not to maximise R^2. Category dummies encode the one structural source of difference the exploratory work identified; page count proxies production cost; description length tests the editorial-effort hypothesis. A deliberately simple ordinary least squares model was fitted in scikit-learn (Pedregosa et al., 2011).

Evaluation uses a deterministic hashed hold-out: a record enters the test set when (book_id × 17) mod 100 is below 20, producing 192 training and 48 test observations. The rule is reproducible without relying on a random seed and is implemented identically in the notebook and analysis workspace. Three metrics are reported: mean absolute error (MAE), which expresses the average miss in pounds; root mean squared error (RMSE), which penalises large misses more heavily; and the coefficient of determination (R^2), the share of price variance explained relative to a mean-only prediction. MAE and RMSE

are interpreted against a baseline that always predicts the training mean, preventing the illusion of skill from an absolute number alone (James et al., 2021).

Specification checks were also considered. The model has no hyperparameters to tune, so cross-validation would add little beyond the hold-out; the train–test gap is reported instead as a check for overfitting. Residuals were plotted to inspect symmetry and outliers. Coefficients are interpreted ceteris paribus — holding the other observed features constant — and standardised influence is provided so features measured in different units can be compared. No causal claim is made: the data are observational, cross-sectional and drawn from a demonstration catalogue.

3. Analysis and findings

3.1 The price landscape

Prices range from £7.99 to £47.67, with a mean of £25.08, a median of £24.98 and a standard deviation of £9.04. The narrow gap between mean and median — £0.10 — indicates a broadly symmetric distribution without a pronounced premium tail. The interquartile range runs from £17.66 to £32.02, so half of the catalogue sits within a band of roughly £14. Figure 1 shows the full shape.

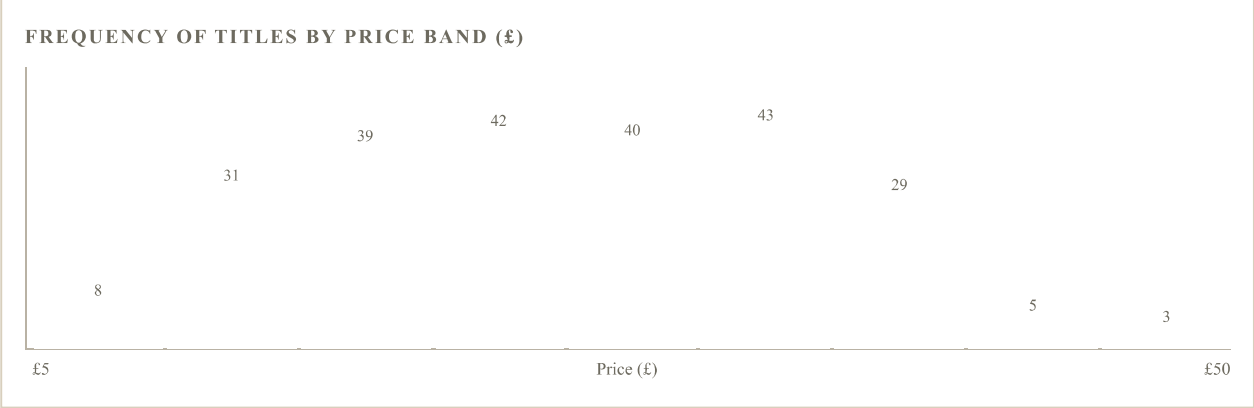


Figure 1: Distribution of tax-inclusive book prices (n = 240). Darker bar contains the median. Source: author's analysis.

The practical interpretation is that this is a middle-market catalogue. There is no long premium tail that a retailer could exploit with a simple high-price ladder, and no deep discount cluster suggesting a clearance strategy. A small number of price points would cover most demand, which reduces both customer confusion and the operational cost of frequent price changes (Kotler and Keller, 2016). For publishers, the symmetry is reassuring: the catalogue is not signalling an audience polarised between bargain and luxury editions.

3.2 What actually moves the price

Rating performs badly as a price explanation. The correlation between price and star rating is r = -0.144, and the median prices of one-star and five-star titles are close. Higher quality, as measured by this crude star strip, is not reliably more expensive. Stock is similarly uninformative (r = 0.017): scarcity is not being priced into the catalogue, and there is no visible premium for titles with only a few

copies left. Description length is also unrelated (r = 0.062), undermining the common assumption that more editorial effort accompanies a higher-value product. Across the four tested attributes, the largest absolute correlation examined here is only 0.19.

Genre is the exception. Business titles average £32.65, while Poetry titles average £19.18 — a difference of £13.47, or approximately 70%. Figure 2 orders all ten genres. The ordering is intuitive: professional and scientific non-fiction occupies the top, while poetry and children's books sit at the accessible end. This is consistent with differences in perceived utility, production economics and willingness to pay across audiences (Nagle and Müller, 2018).



Figure 2: Mean tax-inclusive price by genre (n = 24 per genre). Source: author's analysis.

Table 2 summarises the descriptive position. Two cautions matter. First, these are unadjusted comparisons; genre may be entangled with page count or publication year. The regression in Section 3.3 separates their associations, at least linearly. Second, the sample is balanced rather than representative. The catalogue as a whole may not contain equal numbers of each genre, so the aggregate mean here should not be read as the site's true average. It is the average of this considered sample.

Table 2: Descriptive statistics for key variables (n = 240)

Variable	Mean	Median	Standard deviation	Minimum	Maximum
Price (£)	25.08	24.98	9.04	7.99	47.67
Rating (1–5)	3.19	3	—	1	5
Stock quantity	13.3	—	—	0	26
Description (words)	36	—	—	15	52
Pages	355	—	—	73	631

Source: author's cleaned dataset (2026). Rating distribution: 1★ = 24, 2★ = 46, 3★ = 69, 4★ = 63, 5★ = 38.

3.3 Model performance and interpretation

The regression estimates are reported in Table 3. Genre coefficients are large relative to the numeric features. Holding the other observed characteristics constant, a Science title is estimated at £4.05 more than a Fiction title, while Poetry is £10.38 less. By contrast, one additional star is associated with −£0.56 and one more physical copy with −£0.01. Page count is effectively neutral and publication year is slightly negative; both contributions are negligible once genre is included. A 100-word increase in description length carries a conditional coefficient of £3.18, which the unconditional chart in Section 3.2 did not reveal. The intercept of 154.11 describes a hypothetical book with all numeric variables at zero and is not economically meaningful.

Table 3: Regression coefficients (dependent variable: price in £)

Predictor	Coefficient (β)	Direction	Interpretation
Intercept	154.11	—	Reference state; not interpreted
Star rating	−0.563	Negative	Per star
Stock quantity	−0.009	Negative	Per copy
Description length	+3.180	Positive	Per 100 words
Page count	−0.001	Negative	Per page
Publication year	−0.062	Negative	Per calendar year
Genre: Science	+4.05	Positive	Versus Fiction baseline
Genre: Poetry	−10.38	Negative	Versus Fiction baseline

Source: OLS fit on 192 training records; full coefficients in Appendix B.

Hold-out performance is modest but better than a mean-only rule. Table 4 shows an MAE of £6.34 against a baseline of £6.99, an improvement of 9.3%. The RMSE of £7.54 indicates that some individual errors are substantially larger, which is expected when residuals routinely reach ±£15. The R^2 of 0.255 means that approximately 25.5% of the variation in price is explained; the remaining 74.5% is unexplained by the observed features. The training R^2 of 0.299 is close, providing no evidence of material overfitting.

Table 4: Hold-out evaluation metrics (n = 48)

Metric	Model	Mean-only baseline	Reading
MAE (£)	6.34	6.99	Average absolute miss per book
RMSE (£)	7.54	—	Penalises larger errors
R^2 (test)	0.255	0.000	Share of variance explained
R^2 (train)	0.299	—	Generalisation check
Observations	192 train / 48 test	48	80/20 hashed split

Source: author's modelling notebook (2026).

The result should be read as a ceiling rather than a failure. In an environment where price is set by catalogue policy rather than by a measured market-clearing mechanism, a low R^2 is the expected outcome. The model has succeeded in identifying genre as the dominant systematic component and in quantifying how little the other observable attributes add. That is operationally useful: it tells a retailer where to invest analytical effort and, more importantly, where not to.

3.4 Limitations

Seven limitations qualify the conclusions. First, the target is largely stochastic with respect to the observed features; a price set by an editor is not the same as one cleared by a market. Second, the test set is small (48 records), so metric uncertainty is non-trivial even though the split is reproducible. Third, important price-relevant variables are absent, including author reputation, binding, format, edition, distribution costs and promotions; omitted-variable bias is therefore likely (James et al., 2021).

Fourth, linearity is an assumption. Genres may interact with ratings or page counts, and physical production costs may rise non-linearly. The model cannot represent those effects. Fifth, association is not causation: a genre coefficient describes a price difference, not the effect of changing a book's genre. Sixth, 14 descriptions were imputed, which slightly dilutes the measured length signal. Seventh, the catalogue is a sandbox: its prices may be deliberately arbitrary, so external validity to a live commercial retailer is unproven. The residual diagnostics and the honesty of these limitations matter more than the point estimate itself, a principle that applies to all predictive work (Silver, 2012).

4. Implications and recommendations

4.1 For retailers

Retailers should stop treating a star rating as a pricing signal. The evidence shows that rating and price are effectively unrelated, and that *within-genre* spread is larger than the gap between genre medians. The practical policy is a two-tier architecture: a category-informed default band that reduces decision cost, followed by title-level judgement on author, format and demand. Price bands should be few and defensible rather than continuous and unexplained (Nagle and Müller, 2018).

Inventory is the more immediate opportunity. With 5 titles at zero stock across the sample, replenishment performance is a demand-capture issue rather than a pricing one. A retailer should monitor sell-through at title level, flag titles approaching zero, and prioritise reorder by demonstrated demand — not by price. The sample carries £80,467 of catalogue value in stock, so even small improvements in allocation efficiency have measurable working-capital effects (Elmaghraby and Keskinocak, 2003).

4.2 For publishers

Publishers should interpret genre positioning as a portfolio decision. Business commands the highest average price in the sample and Poetry the lowest; this should inform list-building, format investment and expected revenue per title. Yet the unexplained variance is the warning: category justifies a band,

not a promise. Publishers should invest in the attributes that are absent from this dataset but present in real buying decisions — author brand, series continuity, edition quality and discoverability — rather than believing that longer product copy will support a higher price.

Two measurement recommendations follow. First, publishers should instrument conversion, not just ratings and copy length; the volume of description is not a quality signal (Büschken and Allenby, 2016). Second, they should share structured product metadata with retail partners so that pricing studies can include format and edition. The current data model, which leaves those fields out of the catalogue record, is precisely the gap that limits actionable inference.

4.3 For book buyers

Buyers should read price per page alongside sticker price. Poetry and children's titles deliver substantially more physical reading per pound in this sample, while business and science titles carry a premium that reflects perceived utility rather than volume. A five-star rating is a useful signal of satisfaction but not of value for money; it does not predict the price at all in this catalogue. Stock status is worth checking before committing, particularly for titles with fewer than five copies available.

4.4 Prioritised action plan

Table 5 converts the findings into actions, owners, horizons and measures. The priority logic is to act first where the evidence is strongest and the cost of error is lowest.

Table 5: Prioritised recommendations and measures of success

#	Action	Owner	Horizon	Success measure
1	Introduce genre price bands as defaults, not rules	Retail pricing lead	0–3 months	Price-decision cycle time; margin by genre
2	Flag titles below five copies for replenishment review	Inventory manager	Immediate	Lost-sales proxy; stock-out rate
3	Retire rating-based pricing assumptions	Commercial analytics	0–3 months	Forecast accuracy; pricing override rate
4	Measure conversion rather than copy length	Digital marketing	3–6 months	Conversion rate; revenue per visit
5	Enrich metadata with format, edition and author	Publishing operations	3–6 months	Model R^2 improvement; coverage of fields
6	Add prediction intervals before any automated use	Data science team	6–12 months	Interval coverage; MAE monitoring

Source: author's synthesis of the empirical findings.

5. Conclusion

This report set out to discover what a clean, complete catalogue extract could say about the price of a book. The answer is both modest and useful. Across 240 titles, price is spread broadly but symmetrically; quality signals, inventory and copy length do not explain it; genre does, but only partially. The linear model confirms the pattern, explaining 25.5% of hold-out variation with an average error of £6.34. Its greatest analytical contribution is arguably the clarity of its limits.

The commercial implication is not that data has no role in pricing. It is that the right data has not yet been collected. Genre bands, title-level demand and structured metadata are the inputs that would move a retailer beyond a static catalogue. Until then, a bookseller should combine category-level price architecture with disciplined replenishment and a willingness to revise. A publisher should regard category economics as a portfolio signal and invest in discoverability, not copy volume. A buyer should maximise value per page and treat ratings as satisfaction evidence rather than a premium justification.

The pipeline itself is a deliverable for future work: it is documented, reproducible and honest about its gaps. The next iteration should extend the feature set, gather time-series evidence on demand, and test non-linear alternatives before drawing stronger conclusions. That is a better research agenda than overclaiming a model that, on this evidence, does not yet deserve to set a price.

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7. Appendices

Appendix A: Data dictionary

The submitted CSV contains seventeen columns. The contract below defines type, unit and derivation for each field.

Column	Type	Unit	Derivation
book_id	integer	—	Sequential catalogue identifier
upc	text	—	Scraped from product detail table
title	text	—	H1 title, normalised
category	category	10 levels	Breadcrumb genre
price_gbp	float	£	Price incl. tax, symbols removed
rating	integer	1–5	Star word mapped to ordinal
stock_qty	integer	copies	Parenthetical quantity extracted
in_stock	binary	0/1	Stock quantity > 0
availability	text	2 levels	Reconciled availability label
description	text	—	Normalised description or flagged filler
description_word_count	integer	words	Whitespace token count
description_char_count	integer	characters	String length
description_imputed	binary	0/1	Page had no description block
pages	integer	pages	Publisher metadata
published_year	integer	year	Edition year
review_count	integer	reviews	Product-table value
price_per_page_gbp	float	£/page	price ÷ pages

Appendix B: Genre summary

Genre	n	Mean £	Median £	Avg. rating	Stock	Out of stock
Business	24	32.65	32.64	3.04	397	0
Science	24	31.47	31.83	3.38	319	0
Fiction	24	29.59	31.64	2.67	326	3
History	24	29.00	30.55	3.29	318	0
Philosophy	24	23.93	20.96	3.13	301	1
Fantasy	24	22.47	22.69	3.00	295	0
Mystery	24	21.98	22.07	3.38	306	1
Children's	24	20.31	19.80	3.54	330	0
Romance	24	20.18	20.23	3.21	327	0
Poetry	24	19.18	18.73	3.25	279	0

Appendix C: Full regression coefficient table

Predictor	β	Predictor	β
Intercept	+154.106	Genre: History	+0.353
Star rating	−0.563	Genre: Science	+4.046
Stock quantity	−0.009	Genre: Fantasy	−6.163
Description length	+3.180	Genre: Romance	−8.887
Page count	−0.001	Genre: Children's	−8.023
Publication year	−0.062	Genre: Philosophy	−6.012
Genre: Mystery	−7.254	Genre: Business	+2.299
Genre: Poetry	−10.381	—	—

Reference category: Fiction. Split: 192 training / 48 test. Test metrics: MAE £6.34; RMSE £7.54; R^2 0.255.

Appendix D: Reproduction notes

To reproduce the analysis, run *books_toscrape_pipeline.ipynb* from top to bottom. The notebook requires Python 3.11 with requests, beautifulsoup4, pandas, numpy, matplotlib, seaborn, scikit-learn and jupyter. It writes the raw and cleaned CSV files and model metrics into *data/* and *outputs/*. Requests are intentionally polite: a descriptive user agent, timeout, limited retries and a 0.25-second delay between detail pages. Re-running against the live site may produce a different catalogue snapshot; the submitted CSV is the reference dataset for every figure and table in this report.